

Board notes

Print this and carry it. The deck records what is *projected*; this records what gets *written*, and what gets *typed*.

Four board moments. The ladder, drawn beside the master-equation run — the picture the slide has only as symbols. The exponential, at step 3 of Gillespie. The live code. And the one-line calibration of the variance rule, before the feedback run spends it.

Conventions. **ASK:** put it to the room before writing, and the answer to expect. **POINT:** turn and point at the board rather than the screen. **IF ASKED:** the question that always comes.

Before class, left wing: Thursday's two numbers

Leave these up from the start. The first six surfaces finish Thursday and lean on them.

$$g = \frac{n u^n}{1 + u^n} = 2, \quad u = \frac{x}{K} \quad \alpha_c = \left(\frac{2}{n-2} \right)^{1/n} \frac{n}{n-2}$$

Session 11 wrote this with x , because its x was already in units of K . Today's x is a molecule count, so the u is not decoration: at a mean of 100 with $K = 100$, $u = 1$ and $g = 1.5$ at $n = 3$, where reading x literally would give 3.

Underneath, smaller: **a clock needs gain and delay.**

Right wing, minutes 23–29: the group answers

Draw a square with its diagonal. Write each group's answer **where it moves a point**: shared causes (polymerase, ribosomes, cell size, LacI) along the diagonal; private causes (which polymerase arrived, when an mRNA decayed) across it.

The scatter is on the screen this time, with its two directions marked **A** (along the diagonal) and **B** (across it) and deliberately not named. Write their names when the room supplies them, not before.

ASK at 29, before the sorted surface: *which of these would still be there with a perfect microscope?* All of them — but note what the plot cannot do: uncorrelated read noise in the two channels also spreads points across the diagonal and is counted as intrinsic. Nothing on this scatter separates the two, which is why the noise floor has to be measured separately (their supplement). What the main text does report is the prerequisite for the estimator at all: the two reporters gave statistically equivalent intensity distributions (p. 1184).

Centre, 37–47: the ladder

The run is projected, in two parts: **37–42 builds** the equation and **42–47 solves** it. **Draw the ladder** beside it at the start, because it is a picture and the slide has it as symbols, and leave it up across both parts.

Rungs **0, 1, 2, 3, . . .** stacked vertically. Between each pair, an up-arrow labelled k and a down-arrow labelled γn (with the n of the upper rung). Then write, at part one's step 5:

$$J_n = k P_{n-1} - \gamma n P_n$$

Part one's step 6 is the half-minute that buys the rest. Expand $J_n - J_{n+1}$ on the board and show it reproduces the master equation term for term. The biologists need to see it is the same equation rewritten, not a new one.

Part one ends with J unknown. Before turning the surface, **ASK:** *what could possibly pin that constant down?* Fish for the end of the ladder.

At part two's step 1, POINT at the bottom rung. Nothing below zero, so no traffic can cross the floor: $J_0 = 0$, and steady state has already made every J equal, so every J is zero. **ASK:** *what does $J_n = 0$ let you do with P_n ?* Solve for it from the rung below. Then climb:

$$P_1 = \frac{k}{\gamma} P_0, \quad P_2 = \frac{k}{2\gamma} P_1, \quad P_3 = \frac{k}{3\gamma} P_2 \quad \implies \quad P_n = P_0 \frac{(k/\gamma)^n}{n!}$$

Let the factorial appear on its own. **IF ASKED** why $P_0 = e^{-k/\gamma}$: the probabilities sum to one, and $\sum (k/\gamma)^n / n!$ is the series for $e^{k/\gamma}$.

IF ASKED where the variance comes from without quoting the Poisson — and it is the best question in the hour — multiply the master equation by n and sum over all n . The $\langle n^2 \rangle$ terms cancel between the birth and death sums:

$$\frac{d\langle n \rangle}{dt} = k - \gamma \langle n \rangle$$

which is session 5's ODE, recovered as a statement about an average. Do it again with n^2 and the $\langle n^3 \rangle$ terms cancel, leaving $2\gamma \langle n^2 \rangle = 2k \langle n \rangle + k + \gamma \langle n \rangle$, so $\sigma^2 = k/\gamma$. **Say that the hierarchy closing is special to a LINEAR birth-death** — it is the honest reason the feedback run later has to borrow an approximation. It is worked on the answer sheet.

Centre, at 50: the exponential

At Gillespie step 3, write:

$$\text{Pr}(\text{no event in } \tau) = (1 - a_0 dt)^{\tau/dt} \longrightarrow e^{-a_0 \tau}$$

ASK: *who has seen this before?* Radioactive decay, a Poisson process, a memoryless wait. **Then say the sentence that is the whole explanation:** between events nothing about the system changes, so the chance of something happening in the next instant is the same instant after instant. A process that forgets how long it has already waited can only have an exponential wait. The exponential is not a modelling choice here; it is forced.

Then invert it: $u = e^{-a_0\tau}$, so $\tau = -\ln u/a_0$.

CHECK out loud: a bigger a_0 gives a shorter wait. If the code divides by a_0 where it should, that is what it does.

Live code, at 54: type this

Into a fresh notebook cell, projected. Type it, do not paste it. The room should see it built.

```
import numpy as np
rng = np.random.default_rng(0)
k, gam, T = 10.0, 1.0, 5000.0
t, n = 0.0, 0
ts, ns = [0.0], [0]
while True:
    a = np.array([k, gam * n])
    a0 = a.sum()
    tau = -np.log(rng.random()) / a0
    if t + tau > T:
        break
    t += tau
    n += 1 if rng.random() * a0 < a[0] else -1
    ts.append(t); ns.append(n)
ts, ns = np.array(ts), np.array(ns)
dt = np.diff(np.append(ts, T))          # how long each state was held
keep = ts > 10                          # drop the climb from n = 0
m = np.sum(ns[keep] * dt[keep]) / dt[keep].sum()
v = np.sum((ns[keep] - m) ** 2 * dt[keep]) / dt[keep].sum()
m, v
```

Expect both near 10 (seed 0 gives 10.03 and 9.91, in under a second). **Then run** `ns[keep].mean()` and show it comes out near 10.5 (seed 0: 10.53). **ASK:** *why too high?* A state with more molecules has a higher total rate $k + \gamma n$, so it is left sooner and logged more often per unit time. The event-weighted mean is $\langle n(k + \gamma n) \rangle / \langle k + \gamma n \rangle = 210/20 = 10.5$ exactly.

If late: skip the event-average comparison and say it in one sentence. Do not skip the time weighting.

Centre, at 57: the rule, calibrated

Write the rule before the feedback run spends it, and check it on the case already solved. Define the return rate **first**, in one line, because it is the only new object in the run:

$$\text{kick it by } \Delta : \quad \frac{d\Delta}{dt} = -r \Delta \quad \Rightarrow \quad \Delta \sim e^{-rt}$$

so $1/r$ is how long a fluctuation *lasts*. Then:

$$\sigma^2 = \frac{\text{noise in}}{2 \times \text{return rate}} \quad \text{constitutive:} \quad \frac{k + \gamma n}{2\gamma} = \frac{2k}{2\gamma} = \frac{k}{\gamma} \checkmark$$

Then underneath, for self-repression at the same mean:

$$\frac{2\gamma x}{2\gamma(1+g)} \quad \Rightarrow \quad \text{Fano} = \frac{1}{1+g}$$

POINT at the left wing: that g is Thursday's g .

What to cut if you are late

In this order, last first:

1. The second Becskei surface's arithmetic. Keep the threefold — it is the chromosomal control, which is not the transient one — and keep the sentence that their "stability" is our return rate. Drop the $\sqrt{2}$ comparison.
2. The event-average half of the live code.
3. Item 4 on the handout. Items 1–3 cover T31, T35 and T17 — but item 4 is the only student practice on the sum of squares before PS6, so say the identity out of the deck rather than leaving it stated and unused.

Never cut the sweep surface (PS5 Q5a–b is due Thursday), the ladder, or the NAR run (T17 is owed to PS6 and has no other demonstration).